

AI Sepsis Risk Assessment Report

Generated by Sepsis GenAI Clinical Decision Support System
Date: 2026-03-05 21:45

1. Patient Clinical Data

Vital / Lab	Value
HR	100
Temp	37.40
MAP	75
Resp	18
O2Sat	97.00
Lactate	1.80
Creatinine	1.00
WBC	9.50

2. Model Risk Assessment

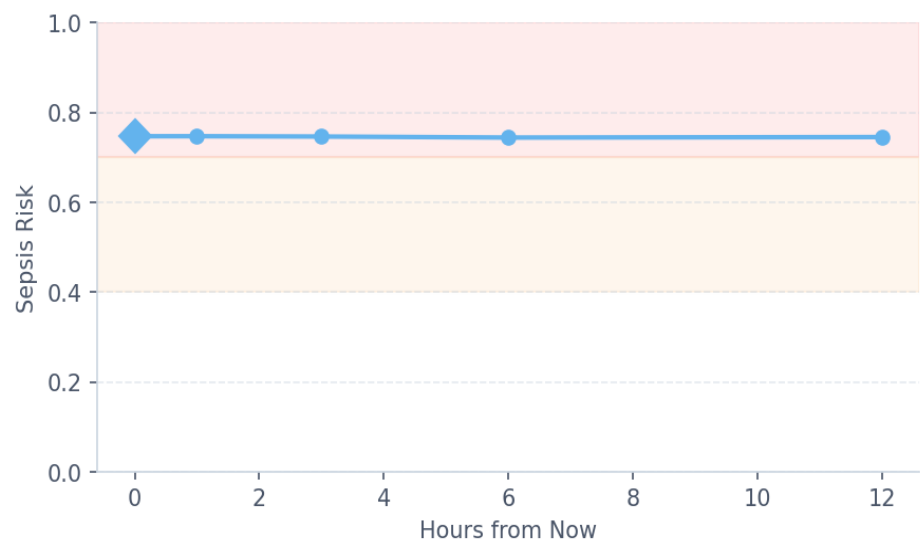
Metric	Value
ML Risk Score	0.071
LSTM Risk Score	0.000
Ensemble Score	0.043 (LOW)
Model Agreement	MEDIUM
Uncertainty Score	0.0710
Clinical Reliability	HIGH

3. Clinical Scoring Systems

Scoring System	Score	Interpretation
qSOFA	0/3	qSOFA score 0/3 — negative screen. No qSOFA criteria currently met.
SOFA	0/24	SOFA score 0/24 — minimal organ dysfunction.

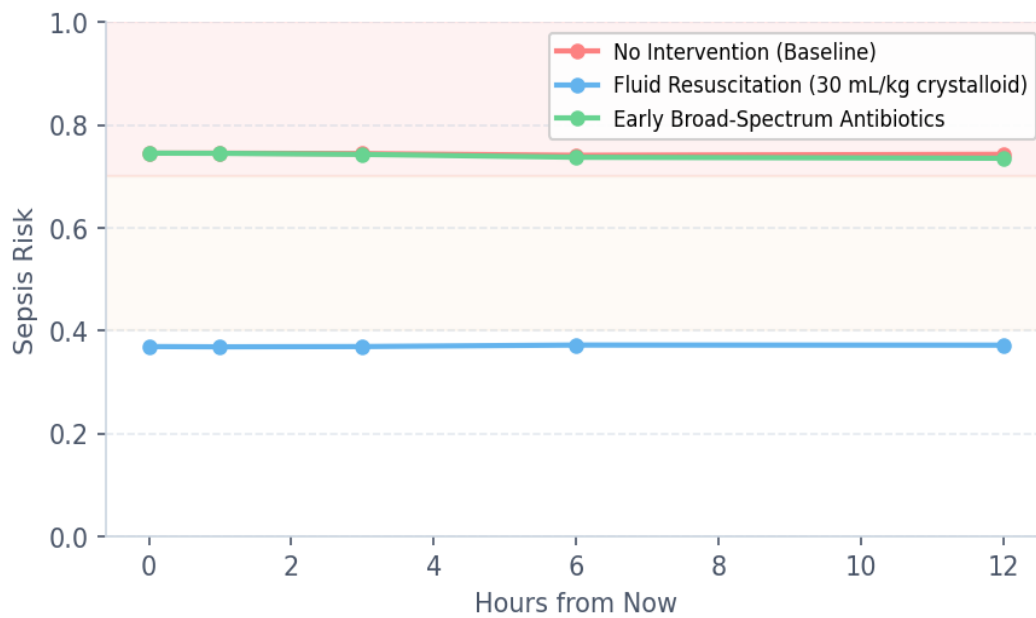
4. Risk Trajectory Forecast

Hour	Predicted Risk
t+1h	0.747
t+3h	0.746
t+6h	0.744
t+12h	0.745



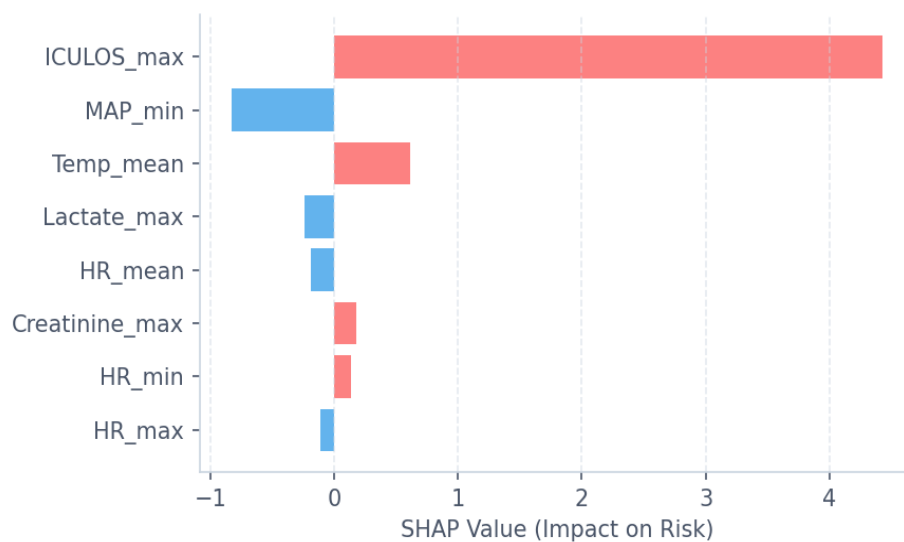
5. Digital Twin Simulation

Intervention	Projected Risk
No Intervention (Baseline)	0.743
Fluid Resuscitation (30 mL/kg crystalloid)	0.372
Early Broad-Spectrum Antibiotics	0.735



6. Feature Importance

Feature	Impact
ICULOS_max	+4.4298 (↑ increases risk)
MAP_min	-0.8282 (↓ decreases risk)
Temp_mean	+0.6084 (↑ increases risk)
Lactate_max	-0.2415 (↓ decreases risk)
HR_mean	-0.1943 (↓ decreases risk)



7. Similar ICU Cases

Patient ID	Outcome	Similarity Score
manual_input	No sepsis diagnosed during ICU stay.	1.000
sim_001	Sepsis diagnosed at hour 18.	0.595
sim_002	No sepsis diagnosed during ICU stay.	0.532
sim_005	No sepsis diagnosed during ICU stay.	0.532
sim_003	Sepsis diagnosed at hour 22.	0.506

8. AI Clinical Reasoning

The ensemble score is 0.0426, which falls below the threshold for moderate risk (0.4-0.7). The patient's vital signs, including heart rate, mean arterial pressure, temperature, and oxygen saturation, are stable and do not indicate organ dysfunction. The lab values, including lactate, creatinine, and white blood cell count, are also stable and do not suggest a dysregulated host response. The qSOFA score is 0/3, indicating no criteria for sepsis are currently met. The SOFA score is 0/24, indicating minimal organ dysfunction. The Sepsis-3 Definition guidelines state that sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection. Given the stable vital signs and lab values, I do not think this patient meets the criteria for sepsis.

This AI-generated report is intended for clinical decision support only and must not replace physician judgement.